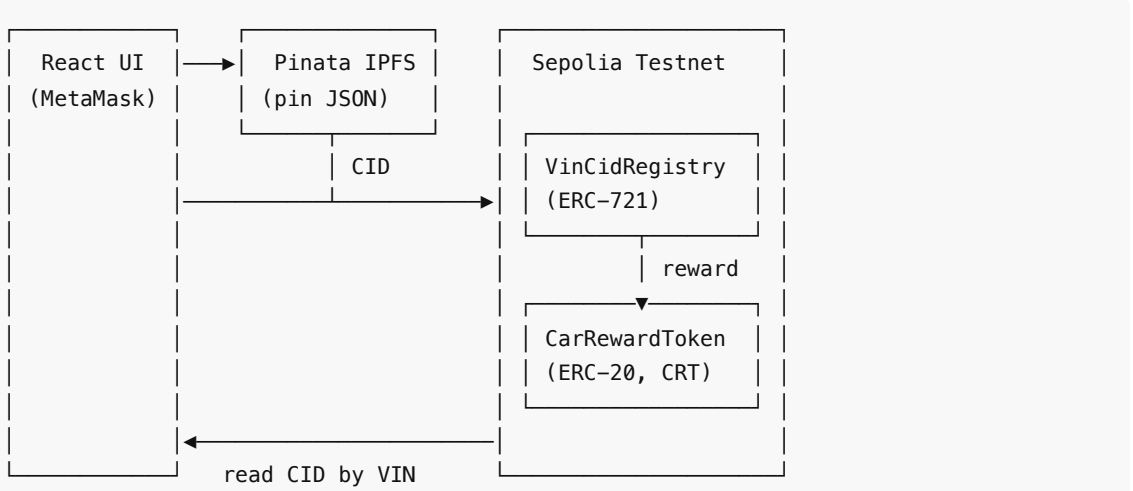
Car Repair NFT

A dApp that records vehicle repair history on-chain. Each car — identified by its 17-character VIN — gets a single NFT whose `tokenURI` points at the latest IPFS CID for that car's repair metadata. Only a designated `minter` address (a registry operator, separate from the contract `owner`) may register new VINs; the NFT is assigned to a `recipient` argument (typically the car owner's wallet). Updating a record rewrites the NFT's URI in place (no new mint), and the registry pays out an ERC-20 reward (CRT) to the recipient on the initial mint.

Flow diagram: https://excalidraw.com/#json=zV5wVQt8GJoK-GYiO-DQn,5mQBcgQrwVfJEm3sxA0Dyw

Architecture



Write flow:

1. The **minter** operator (e.g. a registry admin) connects MetaMask (Sepolia) in the frontend. To register a new VIN, the connected wallet must equal the contract's `minter()`.
2. On submit, the frontend pins a JSON metadata object to IPFS via Pinata and gets back a CID.
3. The frontend calls `VinCidRegistry.storeCid(vin, cid, recipient)`. The first call for a VIN mints its NFT to `recipient` (the car owner). Later calls update the `tokenURI` — updates are open in this POC build (anyone can call them).
4. On mint only, the registry attempts to transfer `rewardAmount` of CRT to the `recipient` (best-effort — silent on failure, so the write still succeeds if funding is empty).
5. To read history, the UI calls `getCidByVin(vin)` and fetches the JSON from `https://gateway.pinata.cloud/ipfs/<cid>`.

Repository layout

```
car_nft/
├── contracts/
│   ├── car_nft_sc.sol          # VinCidRegistry (ERC-721 URI storage)
│   └── car_reward_token.sol    # CarRewardToken (ERC-20, "CRT")
└── frontend/
    ├── public/
    │   ├── index.html
```

```
    │   └── _redirects          # SPA fallback (Netlify-style)
    ├── src/
    │   ├── App.js             # Search + create/update forms
    │   ├── components/
    │   │   └── MetaMaskLogin.jsx
    │   └── utils/
    │       ├── contract_abi.json
    │       ├── contract_utils.js          # chainId → contract address
    │       ├── pinata_ipfs_nft_service.js # IPFS pin + on-chain write
    │       └── validation.js              # VIN / CID / car-data checks
    ├── package.json
    └── vercel.json            # SPA rewrite for Vercel
```

Smart contracts

`VinCidRegistry` (contracts/car_nft_sc.sol)

ERC-721 with URI storage. Token id is `uint256(keccak256(bytes(vin)))`, so each VIN maps to exactly one NFT.

Two roles:

- `owner()` — admin (set at deploy via `Ownable`). Configures reward params and the minter.
- `minter` — registry operator authorized to register new VINs. Set at deploy and changeable by `owner` via `setMinter(address)`. Updates to existing records are open in this POC and do not require either role.

Key functions:

- `storeCid(string vin, string cid, address recipient)` — mint or update. Requires `vin` length 17 and non-empty `cid`. On a new mint, `msg.sender` must be `minter` and `recipient` must be non-zero; the NFT is minted to `recipient` and the CRT reward is paid to `recipient`. On updates, `recipient` is ignored. Emits `CidStored(vin, cid, tokenId)`.
- `minter() → address` / `setMinter(address)` — read or update the minter (`setMinter` is owner-only). Emits `MinterChanged(previousMinter, newMinter)`.
- `getCidByVin(string vin) → string` — latest CID for a VIN.
- `getAllVins() → string[]` / `getAllCidsAsList() → string[]` — enumerate the registry.
- `setRewardToken(address)` / `setRewardAmount(uint256)` — owner-only configuration.
- `withdrawToken(IERC20, address, uint256)` — owner-only; recover ERC-20 funds held by the registry.

`CarRewardToken` (contracts/car_reward_token.sol)

Standard OpenZeppelin ERC-20 named `CarRewardToken` (symbol `CRT`). Mints 1,000,000,000 CRT to the deployer at construction; the owner can `mint(to, amount)` more later.

Deploying the smart contracts

The repo ships a Hardhat setup at the root. Two flavors:

- **Local** — a persistent EVM node on `127.0.0.1:8545` with 20 prefunded test accounts. Fast, free, no faucet needed. Use this for development on the `dev` branch.

- **Sepolia** — Ethereum's public testnet. Requires a Sepolia RPC endpoint, a funded deployer wallet, and (optionally) an Etherscan API key for source verification. Used for production via `main` → Vercel.

Both flavors run the same `scripts/deploy.js`: deploy `CarRewardToken`, then `VinCidRegistry`, then fund the registry with CRT and set the per-mint reward.

Prerequisites

One-time install at the repo root:

```
npm install
```

Scripts available from the repo root (`package.json`):

Command	What it does
`npm run compile`	Compile contracts (output → `artifacts/`)
`npm run node`	Start a persistent local EVM JSON-RPC at `127.0.0.1:8545` (Cancun EVM, chainId `31337`)
`npm run deploy:local`	Deploy to the running local node and auto-write the registry address into frontend/.env.local
`npm run deploy:sepolia`	Deploy to Sepolia using credentials from root `.env`

Configure environment

Copy .env.example to `.env` at the repo root. Leave it empty for local deploys; fill in for Sepolia / verify:

Variable	Required for	Notes
`SEPOLIA_RPC_URL`	Sepolia	Alchemy / Infura / QuickNode endpoint
`DEPLOYER_PRIVATE_KEY`	Sepolia	Dedicated test wallet — never reuse a mainnet key
`ETHERSCAN_API_KEY`	`npx hardhat verify`	Free at https://etherscan.io/myapikey
`INITIAL_MINTER`	optional	Defaults to deployer address
`REWARD_AMOUNT`	optional (default `10`)	CRT per mint
`REWARD_FUND`	optional (default `1000000`)	CRT pool funded into the registry on deploy

`.env` is gitignored. Never commit real values.

Option 1 — Local deploy (Hardhat, no Sepolia)

For day-to-day development. Two terminals from the repo root:

```
# Terminal 1 — leave running
npm run node

# Terminal 2 — deploys CRT + registry, funds it
npm run deploy:local
```

What `deploy:local` does:

1. Deploys `CarRewardToken` and `VinCidRegistry`.
2. Transfers `REWARD_FUND` CRT to the registry and sets `rewardAmount`.
3. Writes a deployment artifact to `deployments/localhost.json` (gitignored, ephemeral).
4. **Auto-populates** `REACT_APP_SMART_CONTRACT_ADDRESS_LOCAL` in frontend/.env.local — no copy-paste needed.

To use it from the frontend:

- In MetaMask, add the localhost network: RPC `http://127.0.0.1:8545`, chainId `31337`, currency `ETH`.
- Import one of the test private keys printed by `npm run node` (account 0 = `0xf39Fd6e5…F92266`, the well-known Hardhat test account).
- Restart `npm start` in `frontend/` so CRA picks up the new `.env.local` value.

State persists for as long as `npm run node` keeps running. Killing the node wipes everything; the next `npm run deploy:local` produces fresh addresses.

> ***Warning**: Hardhat's test mnemonic is public knowledge — anyone running `npx hardhat node` gets the same 20 keys. Never use these accounts on any real network.*

Option 2 — Sepolia deploy

Fill in `.env` first (at least `SEPOLIA_RPC_URL` and `DEPLOYER_PRIVATE_KEY`). The deployer wallet needs ~0.05 Sepolia ETH from a faucet:

- https://www.alchemy.com/faucets/ethereum-sepolia
- https://sepoliafaucet.com

Then:

```
npm run deploy:sepolia
```

What `deploy:sepolia` does:

1. Deploys CRT + registry, funds it (same as local).
2. Writes `deployments/sepolia.json` — **committed** as the team's source of truth for what's currently live.
3. Prints the registry address to paste into Vercel's UI (see Deploying the frontend to Vercel below).

Verifying source on Etherscan (recommended)

Verification unlocks **Read Contract** / **Write Contract** tabs on Etherscan — letting anyone read state and admins run owner-only functions directly from the explorer.

```
# CarRewardToken — no constructor args
npx hardhat verify --network sepolia <crt-address>

# VinCidRegistry — constructor: (rewardTokenAddress, initialMinter)
npx hardhat verify --network sepolia <registry-address> <crt-address> <initial-
minter-address>
```

Re-verifying an already-verified contract is a no-op. Requires `ETHERSCAN_API_KEY` in `.env`.

Roles

Role	Set by	Authority
`owner()`	`Ownable` constructor (= deployer)	Configures `rewardToken`, `rewardAmount`, `minter`. Can `withdrawToken`.
`minter`	`INITIAL_MINTER` env var, or deployer if unset (changeable by `owner` via `setMinter`)	Authorized to register new VINs via `storeCid` (the mint path).
Anyone	—	Can update an existing VIN's CID (POC behavior; lock down later if needed).

Reference deployment (Sepolia)

The latest live addresses are always in deployments/sepolia.json. Inspect on the explorer:

- CarRewardToken on Sepolia Etherscan
- VinCidRegistry on Sepolia Etherscan

After a redeploy, commit the updated `deployments/sepolia.json` and update `REACT_APP_SMART_CONTRACT_ADDRESS` in Vercel's UI (see Vercel section).

Frontend

Stack: React 18 (CRA), Material UI 5, ethers v5, web3 v4, MetaMask provider.

Environment variables

Copy frontend/.env.example to `frontend/.env.local` and fill it in. CRA reads `REACT_APP_*` from `.env.local` at build time and inlines them into the bundle, so **do not put truly secret keys here** — anything `REACT_APP_*` is visible in the browser bundle.

```
REACT_APP_PINATA_API_URL=https://api.pinata.cloud/pinning
REACT_APP_PINATA_JWT=<your-pinata-jwt>
REACT_APP_SMART_CONTRACT_ADDRESS=<sepolia-registry-address>          # leave empty if
not testing against Sepolia from npm start
REACT_APP_SMART_CONTRACT_ADDRESS_LOCAL=<hardhat-registry-address>    # auto-written by
`npm run deploy:local`
```

The chain → contract mapping lives in frontend/src/utils/contract_utils.js: Sepolia (`0xaa36a7`) and Hardhat localhost (`0x7a69`). To support another network, add an entry there.

For production builds on Vercel, **don't ship a `.env.local`** — set the same variables as Project Environment Variables in Vercel's UI instead. See Deploying the frontend to Vercel.

Run locally

```
cd frontend
npm install
npm start            # http://localhost:3000
```

Other scripts: `npm run build`, `npm test`, `npm run clean` (nuke build + node_modules), `npm run fix-install` (cache-clean + reinstall).

Using the app

1. Open the app and click **Connect Wallet**. Approve in MetaMask and switch to Sepolia if prompted. The "Create or Update" panel shows the contract's current `minter` address and whether your connected wallet matches.
2. **Search**: enter a 17-char VIN and click *Load Car NFT*. The app reads the CID from the contract and fetches the metadata from Pinata's gateway. After this call the app knows whether the VIN already exists on-chain (controls which fields/buttons appear next).
3. **Register a new car (mint)**: the connected wallet must be the **minter**. Fill in VIN, **Car Owner Wallet (recipient)**, brand, model, year, issue, repair shop, mileage, then *Register New Car NFT*. The app pins the JSON to IPFS and calls `storeCid(vin, cid, recipient)`. The NFT lands in the recipient's wallet; the CRT reward is also paid to the recipient.
4. **Update an existing car**: any connected wallet works (POC behavior). The Recipient field is hidden because the NFT already exists; the update is applied to that VIN's existing NFT. Click *Submit Repair Update*.
5. After confirmation, the tx hash links to Sepolia Etherscan.

Deploying the frontend to Vercel

The repo contains frontend/vercel.json:

```
{
  "git": { "deploymentEnabled": { "dev": false } },
  "rewrites": [ { "source": "/(.*)", "destination": "/" } ]
}
```

Two things wired here:

1. **Only `main` deploys to Vercel.** The `dev` branch is for local development against a Hardhat node and never goes to Vercel — preview deploys for `dev` would be useless because Vercel can't reach `localhost:8545`.
2. **SPA rewrite** — every path falls back to `/` so deep-links hit `index.html` and React handles routing.

One-time setup (Vercel Dashboard)

1. Push the repo to GitHub.
2. https://vercel.com/new → import the repo.
3. **Root Directory**: `frontend` (the React project lives in a subfolder).

4. **Framework Preset**: Create React App (auto-detected).
5. **Build / Output**: defaults are fine (`npm run build` → `build/`).
6. **Environment Variables** — add to the **Production** scope only:

Variable	Value
`REACT_APP_SMART_CONTRACT_ADDRESS`	Sepolia registry address (from `deployments/sepolia.json`)
`REACT_APP_PINATA_API_URL`	`https://api.pinata.cloud/pinning`
`REACT_APP_PINATA_JWT`	Your Pinata JWT

   Do **not** set `REACT_APP_SMART_CONTRACT_ADDRESS_LOCAL` — production users have no Hardhat node.

7. Click **Deploy**.

Subsequent pushes to `main` redeploy automatically. Pushes to `dev` (or any other branch) do nothing in Vercel.

Re-deploying after a Sepolia redeploy

When the registry address changes (every `npm run deploy:sepolia`):

1. Update Vercel → Project Settings → Environment Variables → `REACT_APP_SMART_CONTRACT_ADDRESS` (Production scope).
2. Trigger a fresh build: push any commit to `main` , or **Deployments → → Redeploy**. CRA only injects env vars at build time — without a fresh build the bundle keeps the old address.
3. Commit the updated `deployments/sepolia.json` so the team has the new addresses too.

Vercel CLI alternative

```
npm i -g vercel
cd frontend
vercel login
vercel              # first run: link project, pick "frontend" as root
vercel env add REACT_APP_SMART_CONTRACT_ADDRESS production
vercel env add REACT_APP_PINATA_JWT production
vercel env add REACT_APP_PINATA_API_URL production
vercel --prod
```

A note on `public/_redirects`

That file is the Netlify equivalent of the rewrite. It's harmless on Vercel — the `vercel.json` rewrites take precedence.

Troubleshooting

- **Blank page / 404 on refresh** — confirm the SPA rewrite in `vercel.json` is active.

- **`Contract address: null`** — `REACT_APP_SMART_CONTRACT_ADDRESS` isn't set for the target environment, or the user is on a chain other than Sepolia.
- **`Pinata 401/403`** — invalid, expired, or under-scoped Pinata JWT. The error surfaces the response body — check it for the exact reason.
- **Reward not received** — registry's CRT balance is empty, or `rewardAmount` is `0`. The write itself still succeeded.
- **Secrets in the bundle** — anything prefixed `REACT_APP_` ships to the browser. Scope the Pinata JWT to pinning only, and rotate if leaked.

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